

Sample Mean

- Why?
- Why not?

Sample Mean:

n samples of the RV X , whose mean $E[X]$ we want to estimate.

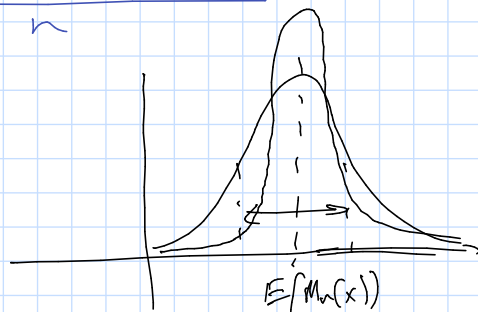
Samples are drawn from the distribution $(f_X(x), F_X(x), P_X(x))$ of RV X and are independent.

n Samples $X_1, X_2, \dots, X_n$.

$$\text{Sample Mean } M_n(X) = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$E[M_n(X)] = E[X]$$

$$\text{Var}[M_n(X)] = \frac{\text{Var}[X]}{n}$$



$$\text{Var}\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right]$$

Chebyshev's Inequality:

For any RV Y ,

$$\begin{aligned} &= \frac{1}{n^2} \text{Var}[X_1 + \dots + X_n] \\ &= \frac{1}{n^2} n \text{Var}[X] = \frac{\text{Var}[X]}{n} \end{aligned}$$

$$P\left[\left|\underbrace{\bar{Y}}_{\text{Sample Mean}} - \underbrace{E[Y]}_{\text{Expected mean value}}\right| \geq \epsilon\right] \leq \frac{\text{Var}[Y]}{\epsilon^2}$$

Sample Mean

Expected
mean value

Exploit And Explore

- Greedy action selection

$$\operatorname{argmax}_{a \in A} Q_t(a)$$

- Introduce exploration

At some rate,
pick/arms other than
from
 $\operatorname{argmax}_{a \in A} Q_t(a)$

Suppose we have m attempts.

I take away $\sum_{i=1}^m R_i$ at the end of m attempts.

Goal:

$$\text{maximize}_{\{a_1, a_2, \dots, a_m\}} E\left[\sum_{i=1}^m R_i\right]$$

$a_1, a_2, \dots, a_m \in A.$

$$A = \{1, 2, \dots, k\}$$

ϵ -greedy

- Probabilistic choice of actions

Exploit
given
current
eval $Q_t(a), \forall a$.

Be greedy

at attempt t .

w.p. $1 - \epsilon$

Explore w.p. ϵ

Pick $\arg\max_{a \in A} Q_t(a)$

$\frac{1}{k}$ $A_{n=1}$
 $\frac{1}{k}$ 2
 $\vdots$
 $\frac{1}{k}$ $A_{n=k}$

$a = \arg\max_{a \in A} Q_t(a)$

$a \neq \arg\max_{a \in A} Q_t(a)$
or uniform

PMF of A_t :

$$P[A_t = a] = \begin{cases} 1 - \epsilon + \frac{\epsilon}{k} \\ \frac{\epsilon}{k} \\ 0 \end{cases}$$

ϵ -greedy

- Think in the long run... after a very large number of arm selections... *(long run, we hope that $Q_t(a) \approx Q^*(a)$)*
- What is the probability that you will select the *optimal* action?

$$P[\text{Optimal action}] = 1 - \epsilon + \frac{\epsilon}{k}$$

ϵ -greedy

- Think in the long run... after a very large number of arm selections...
- What is the long run expected reward? $\leftarrow$

$E[R_t]$ as t becomes very large:

$$E[R_t] = \left(1 - \epsilon + \frac{\epsilon}{k}\right) \phi^* \left(\operatorname{argmax}_a Q_t(a) \right) + \sum_{a \neq \operatorname{argmax}} \frac{\epsilon}{k} \phi^*(a)$$

Problem on Classroom

- On calculating the probability of greedy action selection...

$$\left. \begin{array}{l} Q_1(1) = 4 \\ Q_1(2) = 3 \\ Q_1(3) = 7 \end{array} \right\} \text{Initial values}$$

Suppose we make the following selections & obtain below given rewards.

$$A_1 = 1 \quad A_2 = 2 \quad A_3 = 1 \quad A_4 = 3$$

$$R_1 = 2 \quad R_2 = 1 \quad R_3 = -2 \quad R_4 = 5$$

(i) Calculate the PMF of A_5 , assuming ϵ -greedy action selection. $\epsilon = 0.1$.

$$Q_5(1) = 0$$

$$Q_5(2) = 1$$

$$Q_5(3) = 5$$

$$P(A_5 = a) = \begin{cases} 0.9 + 0.1/3 & a = 3 \\ 0.1/3 & a \in \{1, 2\} \end{cases}$$

Problem on Classroom

- When did we select a greedy action?

Now assume we were following the ϵ -greedy policy starting with action A_1 . When were we greedy & when did we choose exploration?

$Q_1(1) = 4$
 $Q_1(2) = 3$
 $Q_1(3) = 7$

$A_1 = 1$ $A_2 = 2$ $A_3 = 1$ $A_4 = 3$ (can't say)

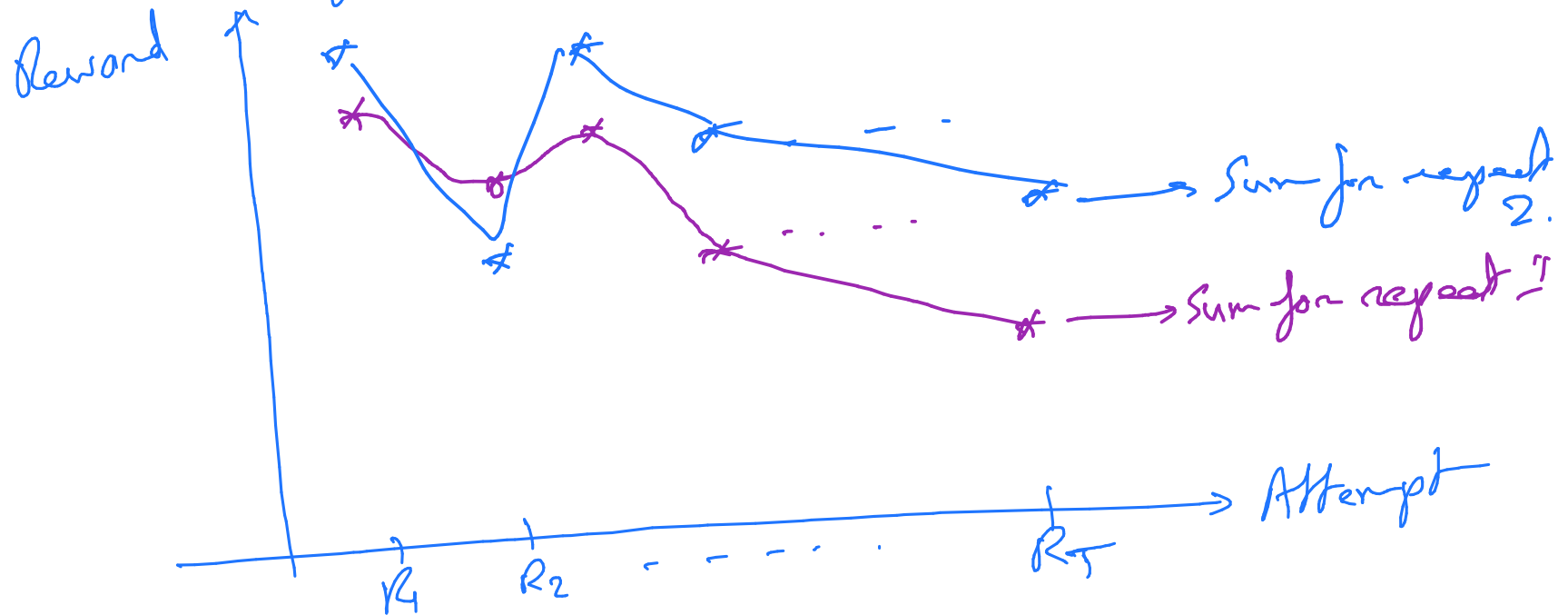
$R_1 = 2$ $R_2 = 1$ $R_3 = -2$ $R_4 = 5$

$Q_2(1) = 2$	$Q_3(1) = 2$	$Q_4(1) = 0$
$Q_2(2) = 3$	$Q_3(2) = 1$	$Q_4(2) = 1$
$Q_2(3) = 7$	$Q_3(3) = 7$	$Q_4(3) = 7$

Simulating Ten Arms

Each arm has a corresponding reward distribution.

- The testbed and what we want to capture
(Simulate N repeats ΔT arm attempts every repeat)



Choice of ε

- 0, 0.01, 0.1
- What will work best in the long run (no constraint on the number of attempts)

Choice of ϵ

- 0, 0.01, 0.1
- What will work best when you have 1000 attempts?

Iterative Calculation (Two Views)

$$Q^{(n)}(a) = \frac{(n-1)}{n} Q^{(n-1)}(a) + \frac{R^{(n)}(a)}{n} \quad \left[\text{Averaging view} \right]$$

$$Q^n(a) = Q^{(n-1)}(a) + \frac{1}{n} (R^{(n)}(a) - Q^{(n-1)}(a))$$

Error in $Q^{(n-1)}(a)$
 w.r.t our current
 value $R^{(n)}(a)$, which is
 an instantaneous (single sample)
 estimate of $Q^*(a)$.

$$Q^{(n-1)}(a) + \underbrace{\alpha_n (R^{(n)}(a) - Q^{(n-1)}(a))}_{\substack{\uparrow \\ \text{Step-size parameter} \rightarrow 0}}$$

sign of the
 error
 gives the
 direction
 in which
 we must
 move
 $Q^{(n-1)}(a)$.

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 Step-size parameter $\rightarrow 0$

Sign of the
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 $Q^{(n-1)}(a)$.

We are reducing/increasing $Q^{(n-1)}(a)$, depending on the
 direction of the update. Given the direction, the amount of change
 in $Q^{(n-1)}(a)$ is a function of α_n & the magnitude of the error.